

All-in-One Scientific Calculator – Project Documentation

1. INTRODUCTION

The All-in-One Calculator is a beginner-friendly Python project developed as our first major step into programming. The aim of this project is to build a single calculator application that can perform multiple types of calculations instead of creating separate programs for each task.

This calculator is capable of handling basic arithmetic operations such as addition, subtraction, multiplication, division, powers, and square roots. In addition, it also includes scientific features like trigonometric functions (sine, cosine, and tangent), logarithmic functions, and health-related calculations such as Body Mass Index (BMI) and Body Fat Percentage.

2. METHODOLOGY

The project follows a modular programming approach. This means that each calculation is written as a separate function. For example, addition, subtraction, trigonometric calculations, and health calculations all have their own functions. This approach makes the program easier to read, understand, test, and improve.

The program uses a menu-driven system where the user is shown a list of available operations. The user selects an option by entering a number, and the program then asks for the required inputs. Based on the user's choice, the corresponding function is executed, and the result is displayed.

Important programming concepts used in this project include:

- Functions for organized code structure
- Loops to keep the program running until the user exits
- Conditional statements to handle choices and errors
- Dictionary-based menu system for easy expansion

Instead of relying on advanced Python libraries, mathematical logic such as Taylor series for sine and cosine functions and series expansion for logarithms was implemented manually. This helped strengthen our understanding of both mathematics and programming fundamentals.

Logic Flow / Pseudocode

START

WHILE True

 DISPLAY menu

 INPUT user_choice

 IF user_choice == 0

 PRINT "Thank you"

 EXIT loop

 IF user_choice not in menu

 PRINT "Invalid option"

 CONTINUE

GET function and required arguments

IF BMI selected

 INPUT weight, height

 CALCULATE BMI

 DISPLAY BMI and status

ELSE IF Body Fat selected

 INPUT gender, measurements

 CALCULATE body fat %

 DISPLAY result and status

ELSE

INPUT required values

CALL selected function

DISPLAY result

END WHILE

END

3. FEATURES OF THE CALCULATOR

The All-in-One Calculator includes the following features:

1. Basic Arithmetic Operations:

- Addition, subtraction, multiplication, and division
- Error handling for division by zero
- Power and square root calculations

2. Trigonometric Functions:

- Sine, cosine, and tangent functions
- Accepts angles in degrees
- Handles special angles such as 0° , 90° , 180° , and 270°

3. Logarithmic Functions:

- Natural logarithm (ln)
- Logarithm base 10
- Input validation for invalid values

4. Health Calculations:

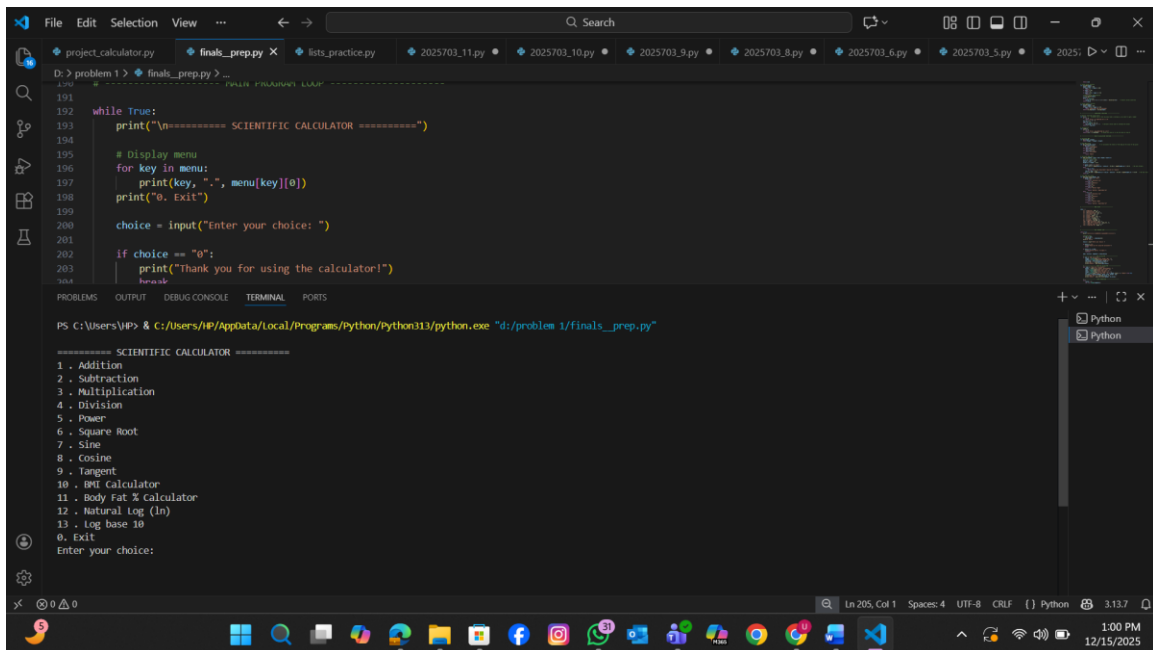
- Body Mass Index (BMI) calculator with health status
- Body Fat Percentage calculator based on gender
- Provides meaningful health category results

All these features are combined into one program, making it practical, interactive, and easy to use.

4. RESULTS AND OUTPUT

The program runs successfully in the Python environment and provides accurate results for all supported calculations. The menu-based interface makes it simple for users to navigate through different options.

For arithmetic and scientific calculations, the program returns precise numerical values. For health-related calculations, the program not only displays numerical results but also provides meaningful interpretations such as health status categories.



```
191
192 while True:
193     print("\n===== SCIENTIFIC CALCULATOR =====")
194
195     # Display menu
196     for key in menu:
197         print(key, "-", menu[key][0])
198     print("0. Exit")
199
200     choice = input("Enter your choice: ")
201
202     if choice == "0":
203         print("Thank you for using the calculator!")
204         break
```

```
PS C:\Users\VIP> & C:\Users\VIP\AppData\Local\Programs\Python\Python113\python.exe "d:/problem 1/finals_prep.py"
===== SCIENTIFIC CALCULATOR =====
1 . Addition
2 . Subtraction
3 . Multiplication
4 . Division
5 . Power
6 . Square Root
7 . Sine
8 . Cosine
9 . Tangent
10 . BMI Calculator
11 . Body Fat % Calculator
12 . Natural Log (ln)
13 . Log base 10
0. Exit
Enter your choice:
```

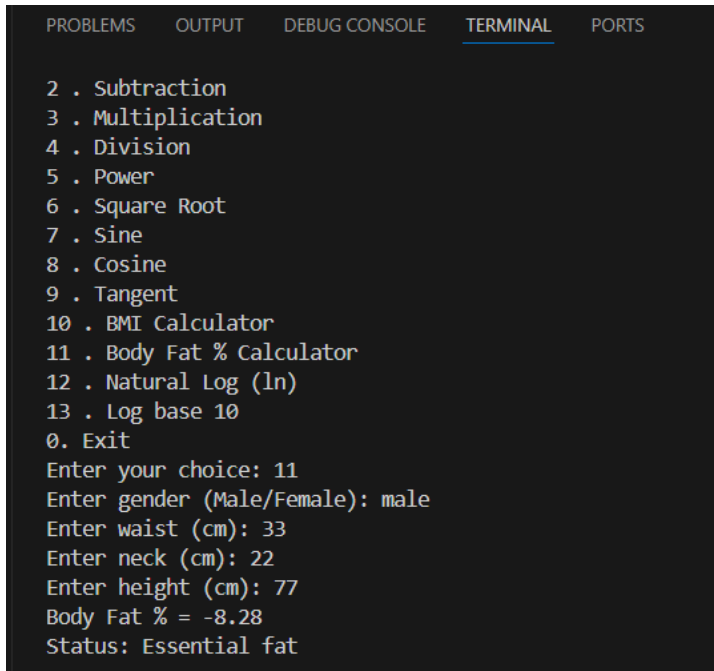
5. CONCLUSION

This project was an excellent learning experience and a strong introduction to Python programming. As our first Python project, it helped us understand how functions, loops, and conditional statements work together to form a complete program.

Some challenges included handling incorrect user inputs and debugging logical errors, especially in loops and mathematical calculations. However, overcoming these challenges improved our problem-solving skills.

Overall, the All-in-One Calculator successfully demonstrates our understanding of Python basics and shows how programming can be used to create useful real-world applications.

6. APPENDIX



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

2 . Subtraction
3 . Multiplication
4 . Division
5 . Power
6 . Square Root
7 . Sine
8 . Cosine
9 . Tangent
10 . BMI Calculator
11 . Body Fat % Calculator
12 . Natural Log (ln)
13 . Log base 10
0. Exit
Enter your choice: 11
Enter gender (Male/Female): male
Enter waist (cm): 33
Enter neck (cm): 22
Enter height (cm): 77
Body Fat % = -8.28
Status: Essential fat
```